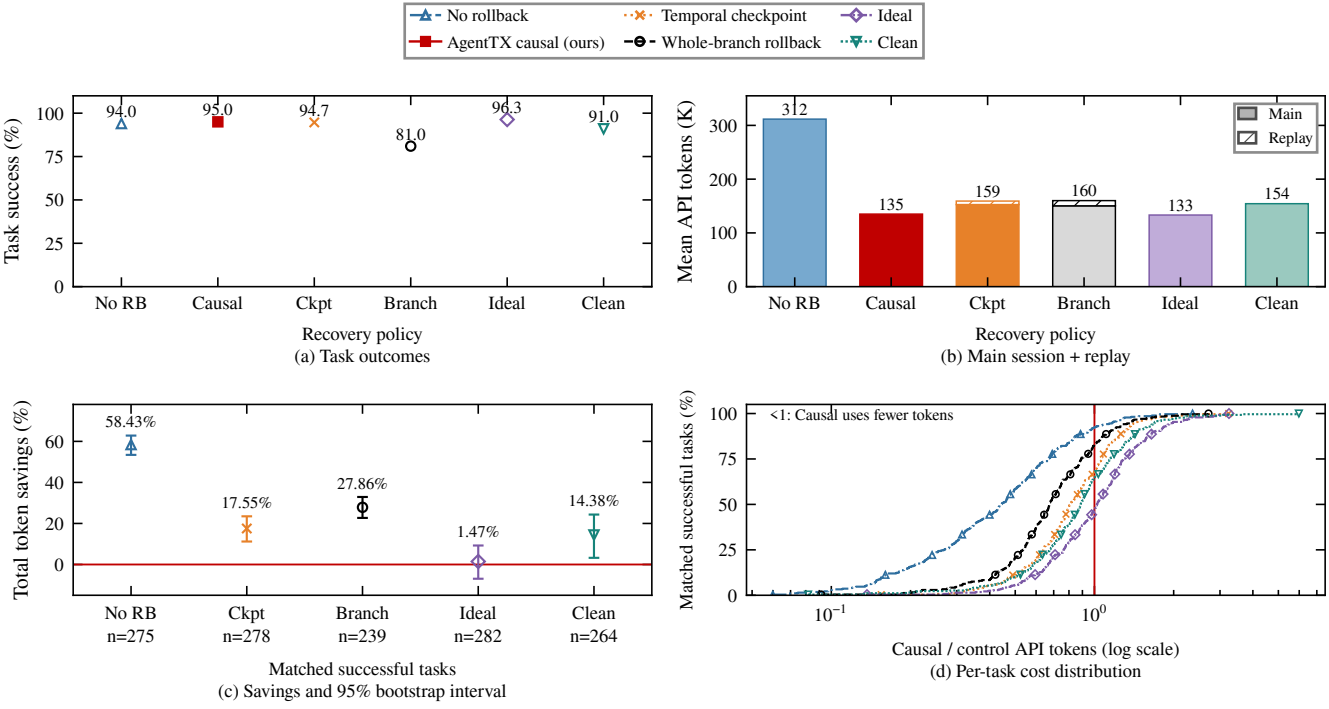
No rollback
Temporal checkpoint
Ideal
AgentTX causal (ours)
Whole-branch rollback
Clean
Task success (%)
94.0
95.0
94.7
81.0
96.3
91.0
No RB
Causal
Ckpt
Branch
Ideal
Clean
Recovery policy
(a) Task outcomes
Mean API tokens (K)
312
135
159
160
133
154
Main
Replay
Recovery policy
(b) Main session + replay
Total token savings (%)
58.43%
17.55%
27.86%
1.47%
14.38%
No RB n=275
Ckpt n=278
Branch n=239
Ideal n=282
Clean n=264
Matched successful tasks
(c) Savings and 95% bootstrap interval
Matched successful tasks (%)
<1: Causal uses fewer tokens
Causal / control API tokens (log scale)
(d) Per-task cost distribution